Jerry Xizhao Fu

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Education

2025–(2027) Collège Jean-de-Brébeuf – IBDP + DEC Pure and Applied Sciences (pre-uni)
2020–2025 Collège Regina Assumpta – Natural Sciences and Robotics

Research Experience

2025–present **CHUM Research Center**

- Processed lung ultrasound elastography sequences, segmented the pleural line, and ran strain analysis in MATLAB to study pleural strain's relation to tidal volume.
- Calibrated spirometry equipment and tuned ultrasound parameters for a biplanar-probe study.
- Observed cardiac surgery, intubation, renal replacement therapy, and more in the OR and ICU.

Publications/Abstracts

- Sep 2025 · CHUM Research Center *Relation entre la déformation pleurale régionale mesurée par élastographie pulmonaire et le volume courant chez des volontaires sains.* S. Chraïbi, K. El Berhoumi, A. Danatoiu, **J. Fu**, P. Vianna, M.H. Roy Cardinal, G. Cloutier, M. Girard (presenting author). Annual Congress of the French Society of Anesthesia and Critical Care (SFAR), Paris, France.

Work Experience

2023–present **DODO Wheels / PublicoTires – Software Developer**

- Collaborating with a small team to build a centralized operations platform for the company.
- Integrated the company website with QuickBooks Online, enabling programmatic management of invoices, customers, and inventory.
- Automated the migration of 3,000+ products and images to the new system via a web scraper.
- 2024 Developed the Doublestar Tire Canada website — doublestartyre.ca

2022–present **FIRST Robotics Competition – Mentor & former Student of Team 3990**

- Created coursework and taught programming, computer vision and AI/ML to junior team members.
- Developed autonomous behaviors and computer vision object tracking on the robot.
- Multiple consecutive Regional Winners; 2024 World Championship Impact Award finalist.

Extracurriculars/Notable Projects

- 2026 **WRCC Brain–Computer Interface Competition (by Tsinghua University) – Grand Prize winner**. Built an EEG motor-imagery classifier using a Riemannian geometry pipeline (4 channels), averaging 77% accuracy (48.6% baseline) across four binary tasks.
- 2026 **TechNexus** – Companion app for FIRST Robotics events, built with two teammates and released in October for iOS, Android, and Windows. Uses Kotlin Multiplatform and SwiftUI. Supports Dynamic Island, Live Activities, and push notifications — technexus.jerryxf.net
- 2025–present **Medical Inference via Vector Embeddings** – An AI system that generates a differential diagnosis from patient-reported symptoms, fusing a biomedical language model with UMLS medical concepts. IB Extended Essay in Computer Science.
- 2026 **HiddenGarden** – Built the storefront for a local florist business, supporting Shopify's Storefront API for checkout — hiddengarden.pages.dev
- 2025–2026 **ORCA** – Designed a real-time intelligence system that fuses live telemetry, sensor and radio data. Prepared a TRL-3 proposal for the Canadian DND/CAF IDEaS program.

- 2024 & 2025 ***Secondaire en spectacle*** – Performed piano-comedy skits with a friend, 400+ audience. Local winner.

Volunteer Experience

- 2025–(2027) Mentoring the new CRC Robotics team at Brébeuf. Rookie Team of the Year award.
- 2026 Played music with a student volunteer group for patients and families at CHU Sainte-Justine.
- 2025 Contributed to the organization of Team 3990's in-house off-season robotics competitions.
- 2025 Guided incoming students as a peer mentor (*Anges gardiens*) at Collège Regina Assumpta.

Software Skills

- **Languages**: TypeScript/JS, Python, Java, Kotlin, Swift, MATLAB
- **Libraries**: React, Express.js, PyTorch, OpenCV, WPILib, Kotlin Multiplatform, SwiftUI
- **Infrastructure**: Docker, Cloudflare, GCP, OAuth, QuickBooks Online

Other Skills

- Machining and fabrication, 3D printing, laser cutting, CNC, lathe; soldering and crimping
- Basic CAD (Onshape, Autodesk Inventor)
- Photography and Adobe Lightroom

Certifications

- **Research ethics**: Good Clinical Practice (CITI), Éthique de la recherche (MSSS), Responsible Conduct of Research (CRCHUM)
- **Canadian Red Cross**: Standard First Aid & CPR/AED Level C, Psychological First Aid